

CS & IT ENGINEERING

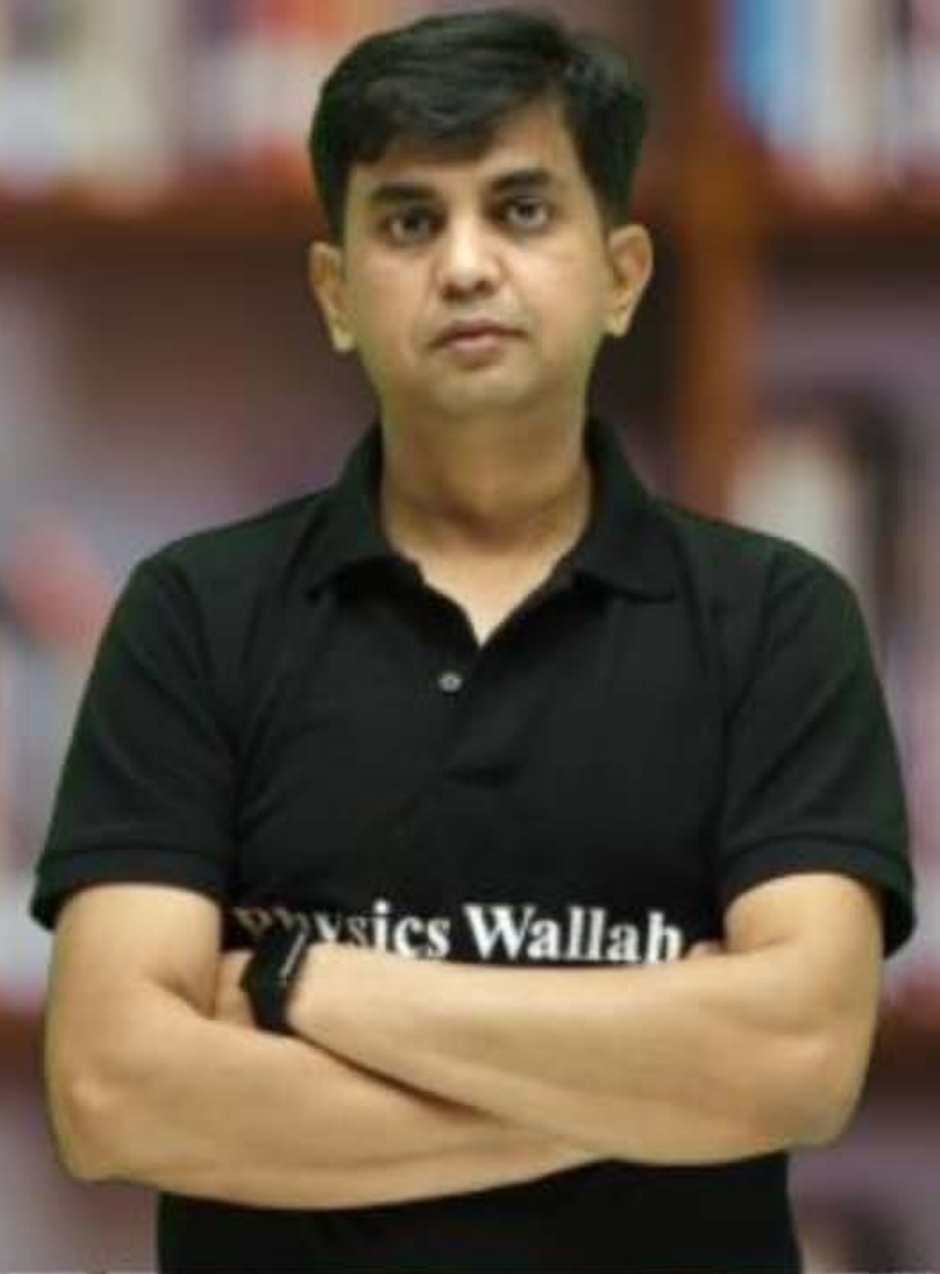


Computer Network

IPv4 Header

Lecture No. - 05

By - Abhishek Sir





Recap of Previous Lecture



Topic

Fragmentation & Reassembly





Topics to be Covered



Topic

Fragmentation & Reassembly



ABOUT ME



Hello, I'm **Abhishek**

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[NAT]

[GATE-2018][2 Mark]



#Q. Consider an IP packet with a length of 4,500 bytes that includes a 20-byte IPv4 header and a 40-byte TCP header. The packet is forwarded to an IPv4 router that supports a Maximum Transmission Unit (MTU) of 600 bytes. Assume that the length of the IP header in all the outgoing fragments of this packet is 20 bytes. Assume that the fragmentation offset value stored in the first fragment is 0.

The fragmentation offset value stored in the third fragment is _____.

Ans = 144

Total Length = [4500 bytes]

Header Size = [20 bytes]

Old Payload Size = [Total Length - Header Size] bytes
 $= (4500 - 20) \text{ bytes} = 4480 \text{ bytes}$

MTU = 600 bytes

New Payload Size = [MTU - Header Size] bytes
 $= (600 - 20) \text{ bytes}$
 $= 580 \text{ byte}$
 $= 576 \text{ byte (if Mbit is one)}$

$$\underbrace{\text{Offset value of the 3}^{\text{rd}} \text{ IP fragment}}_{\uparrow\uparrow} = \text{Old offset value} + \left[\frac{2 * \text{New Payload Size}}{8} \right]$$

$$= 0 + \left[\frac{2 * 576}{8} \right]$$

$$= 0 + 144$$

$$= 144$$

[NAT]

[GATE-2024][2 Mark]



#Q. Consider sending an IP datagram of size 1420 bytes (including 20 bytes of IP header) from a sender to a receiver over a path of two links with a router between them. The first link (sender to router) has an MTU (Maximum Transmission Unit) size of 542 bytes, while the second link (router to receiver) has an MTU size of 360 bytes. The number of fragments that would be delivered at the receiver is _____.

Ans = 6



$$\text{Total Length} = [1420 \text{ bytes}]$$

$$\text{Header Size} = [20 \text{ bytes}]$$

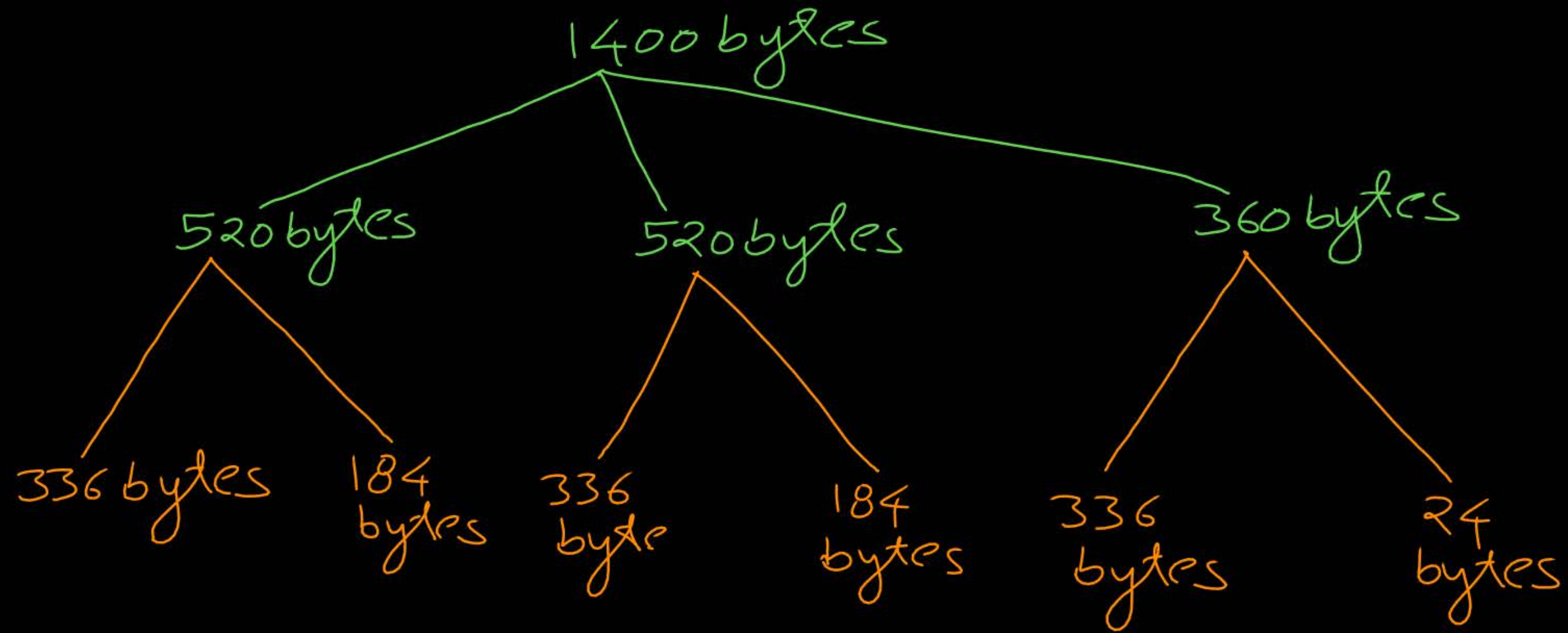
$$\begin{aligned} \text{Old Payload Size} &= [\text{Total Length} - \text{Header Size}] \text{ bytes} \\ &= (1420 - 20) \text{ bytes} = 1400 \text{ bytes} \end{aligned}$$

$$\text{MTU}_1 = 542 \text{ bytes}$$

$$\begin{aligned} \text{New Payload Size}_1 &= [\text{MTU} - \text{Header Size}] \text{ bytes} \\ &= (542 - 20) \text{ bytes} \\ &= 522 \text{ byte} \approx 520 \text{ bytes} \end{aligned}$$

$$\text{MTU}_2 = 360 \text{ bytes}$$

$$\begin{aligned} \text{New Payload Size}_2 &= [\text{MTU} - \text{Header Size}] \text{ bytes} \\ &= (360 - 20) = 340 \text{ bytes} \approx 336 \text{ bytes} \end{aligned}$$

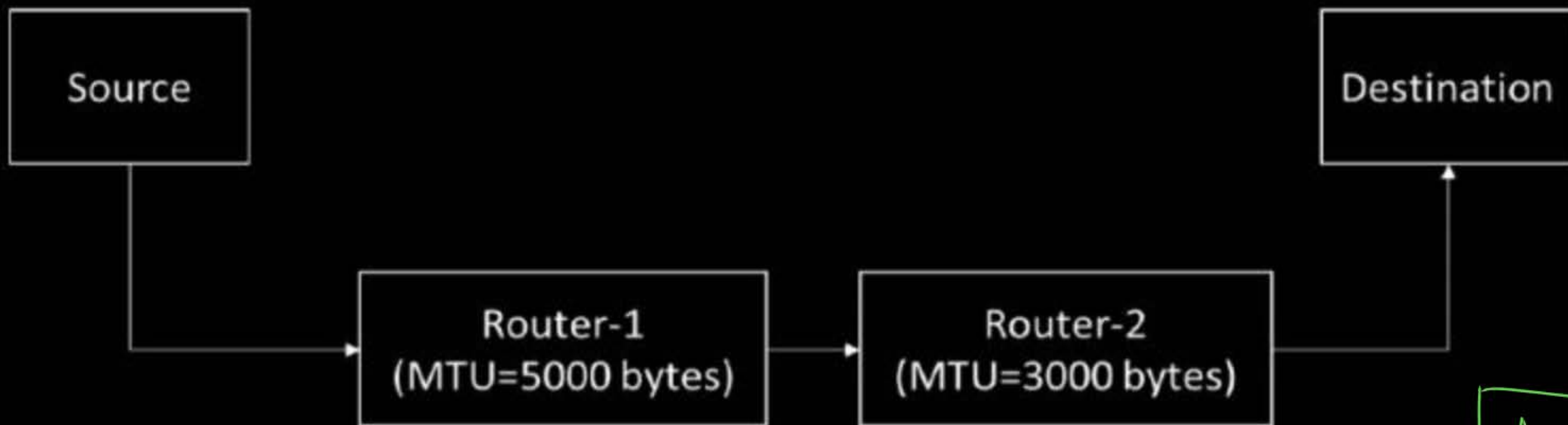


[NAT]

[GATE-2025][2 Mark]



#Q. Suppose a message of size 15000 bytes is transmitted from a source to a destination using IPv4 protocol via two routers as shown in the figure. Each router has a defined maximum transmission unit (MTU) as shown in the figure, including IP header. The number of fragments that will be delivered to the destination is _____. (Answer in integer)



Ans = 7

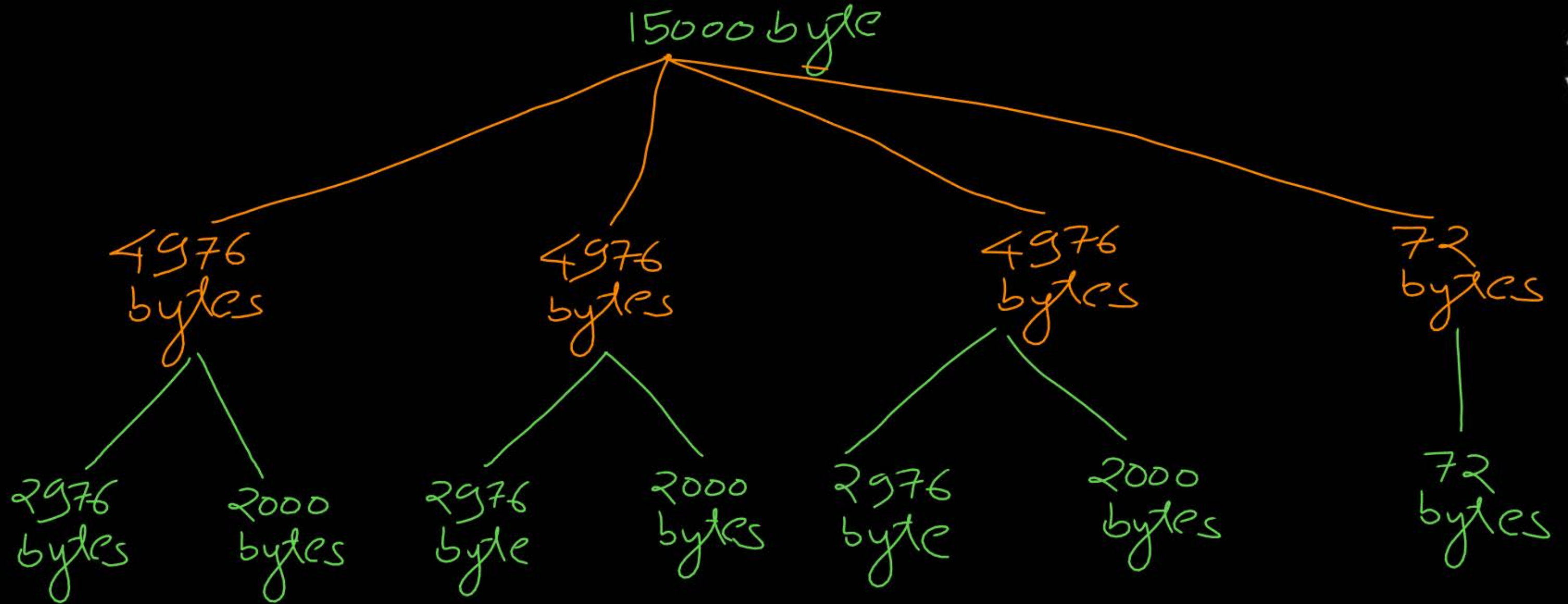
$$\text{Message Size} = 15000 \text{ bytes}$$

$$\text{MTU}_1 = 5000 \text{ bytes}$$

$$\begin{aligned} \text{New Payload Size}_1 &= [\text{MTU} - \text{Base Header Size}] \text{ bytes} \\ &= (5000 - 20) \text{ bytes} \\ &= 4980 \text{ bytes} \approx 4976 \text{ bytes} \end{aligned}$$

$$\text{MTU}_2 = 3000 \text{ bytes}$$

$$\begin{aligned} \text{New Payload Size}_2 &= [\text{MTU} - \text{Base Header Size}] \text{ bytes} \\ &= (3000 - 20) \text{ bytes} \\ &= 2980 \text{ bytes} \approx 2976 \text{ bytes} \end{aligned}$$



Example 10 :-

[NAT]



#Q. Consider UDP segment of size 380 bytes is passed to IPv4 protocol for delivery. MTU for source network is 400 bytes and IPv4 header size is 20 bytes. Calculate total number of IPv4 fragments required to carry the UDP segment after fragmentation?

Ans = 1

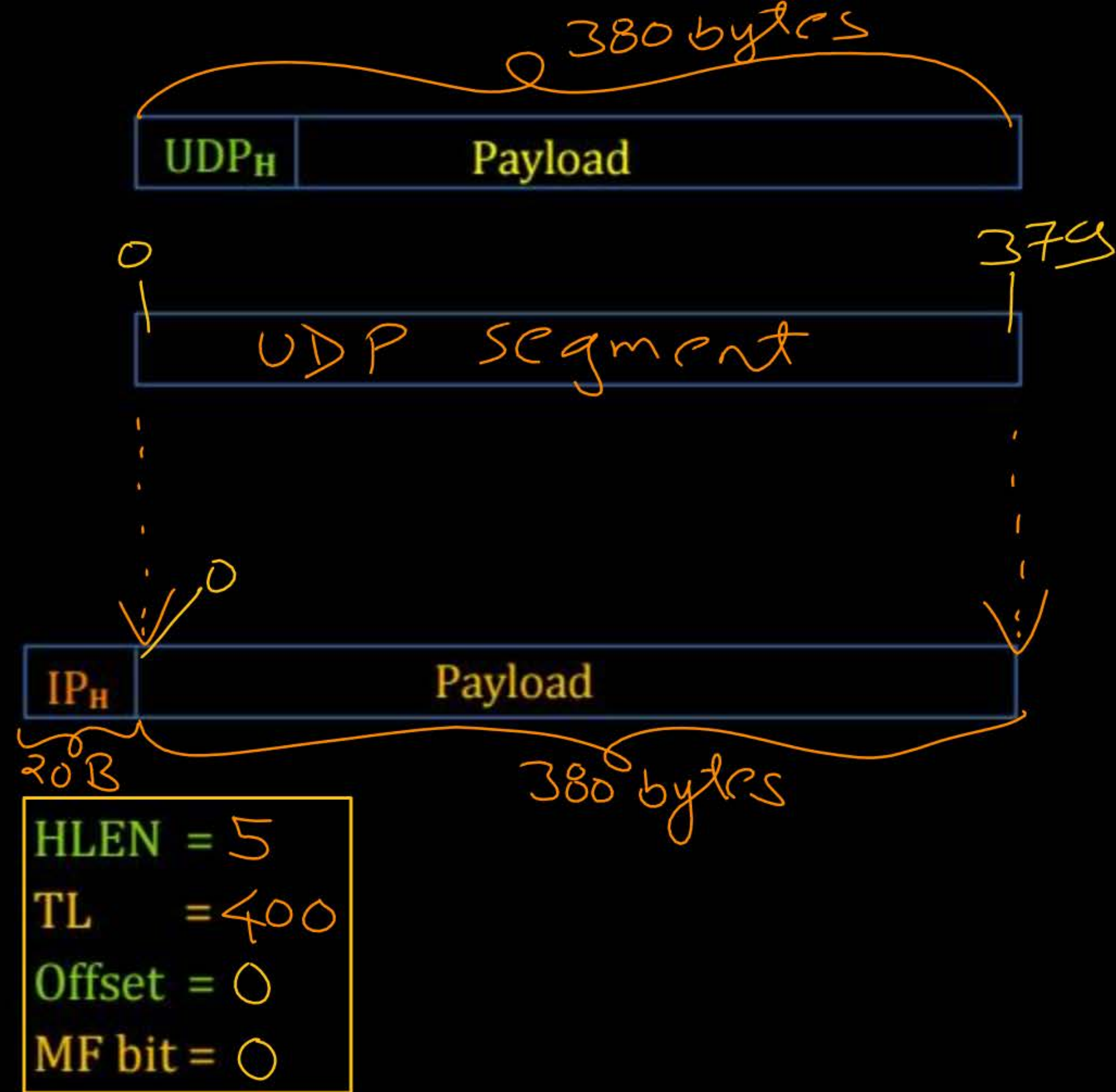
$$\text{UDP Segment Size} = 380 \text{ bytes}$$

$$\begin{aligned} \text{MTU} &= [400 \text{ bytes}] \\ \text{IPv4 Header Size} &= [20 \text{ bytes}] \end{aligned}$$

$$\text{Maximum Payload Size} = [\text{MTU} - \text{Header Size}] \text{ bytes}$$

$$\begin{aligned} &= [400 - 20] \text{ bytes} \\ &= 380 \text{ bytes} \end{aligned}$$

$$\approx 376 \text{ bytes} [\text{when M bit is one}]$$



Example 11 :-

[NAT]



#Q. Consider UDP segment of size 381 bytes is passed to IPv4 protocol for delivery. MTU for source network is 400 bytes and IPv4 header size is 20 bytes. Calculate total number of IPv4 fragments required to carry the UDP segment after fragmentation ?

Ans = 2

$$\text{UDP Segment Size} = 381 \text{ bytes}$$

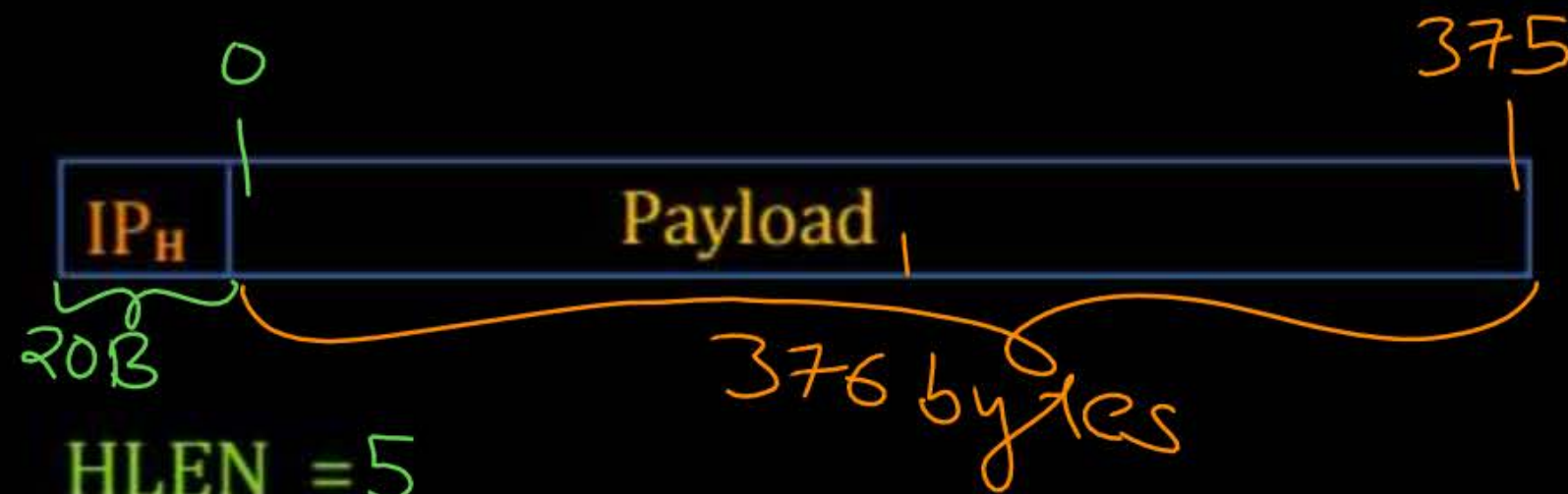
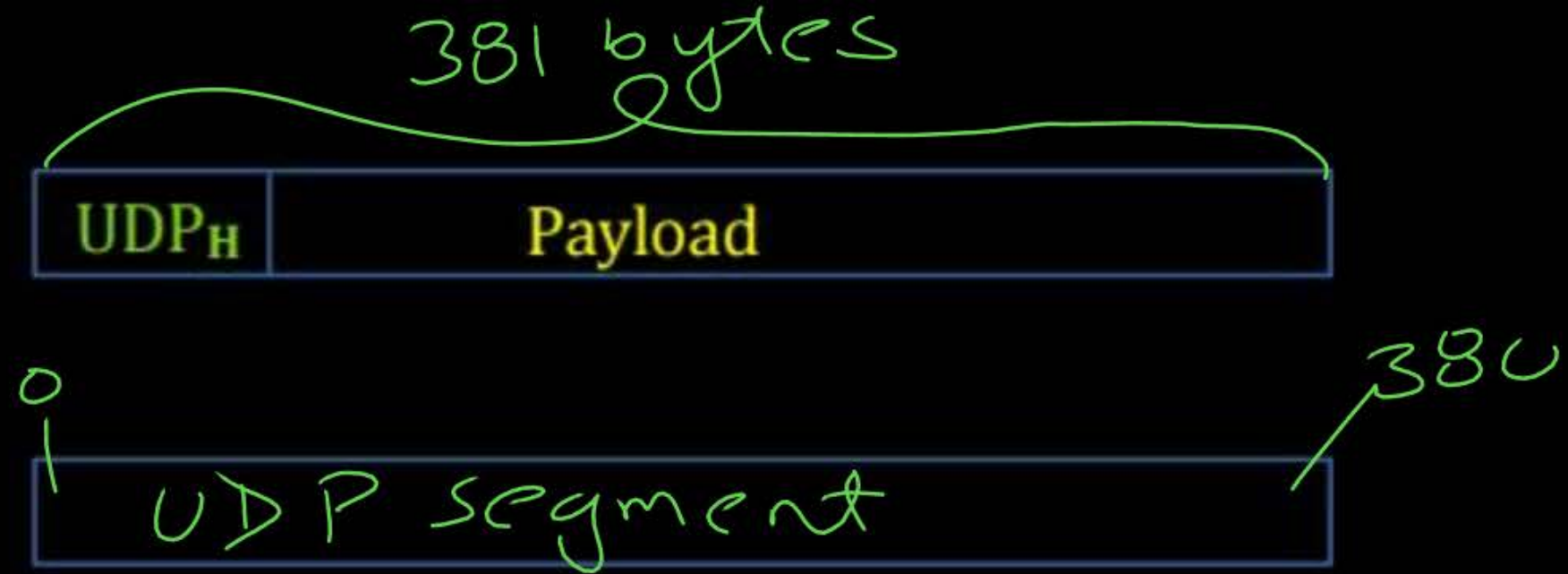
$$\begin{aligned} \text{MTU} &= [400 \text{ bytes}] \\ \text{IPv4 Header Size} &= [20 \text{ bytes}] \end{aligned}$$

$$\text{Maximum Payload Size} = [\text{MTU} - \text{Header Size}] \text{ bytes}$$

$$= (400 - 20) \text{ bytes}$$

$$= 380 \text{ bytes}$$

$$376 \text{ bytes (when } M=1)$$

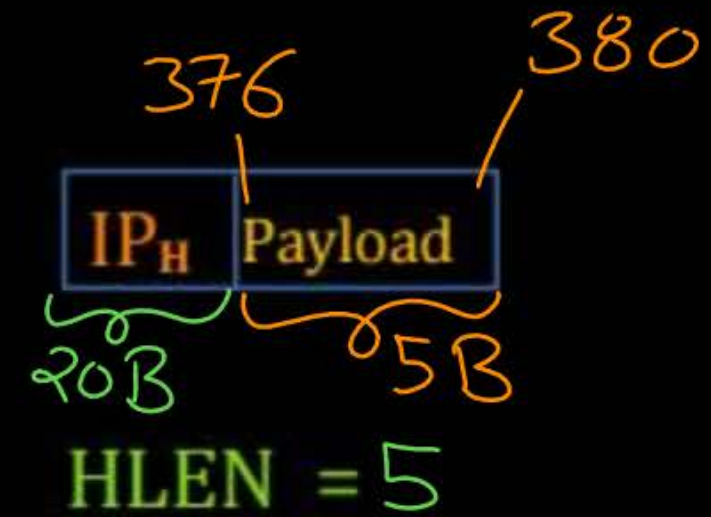


HLEN = 5

TL = 396

Offset = 0

MF bit = 1



HLEN = 5

TL = 25

Offset = 47

MF bit = 0

Example 12 :-

[NAT]



#Q. Consider UDP segment of size 532 bytes is passed to IPv4 protocol for delivery. MTU for source network is 200 bytes and IPv4 header size is 20 bytes. Calculate total number of IPv4 fragments required to carry the UDP segment after fragmentation ?

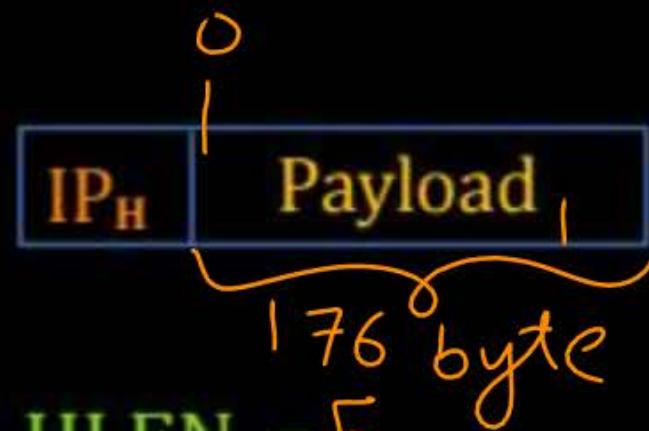
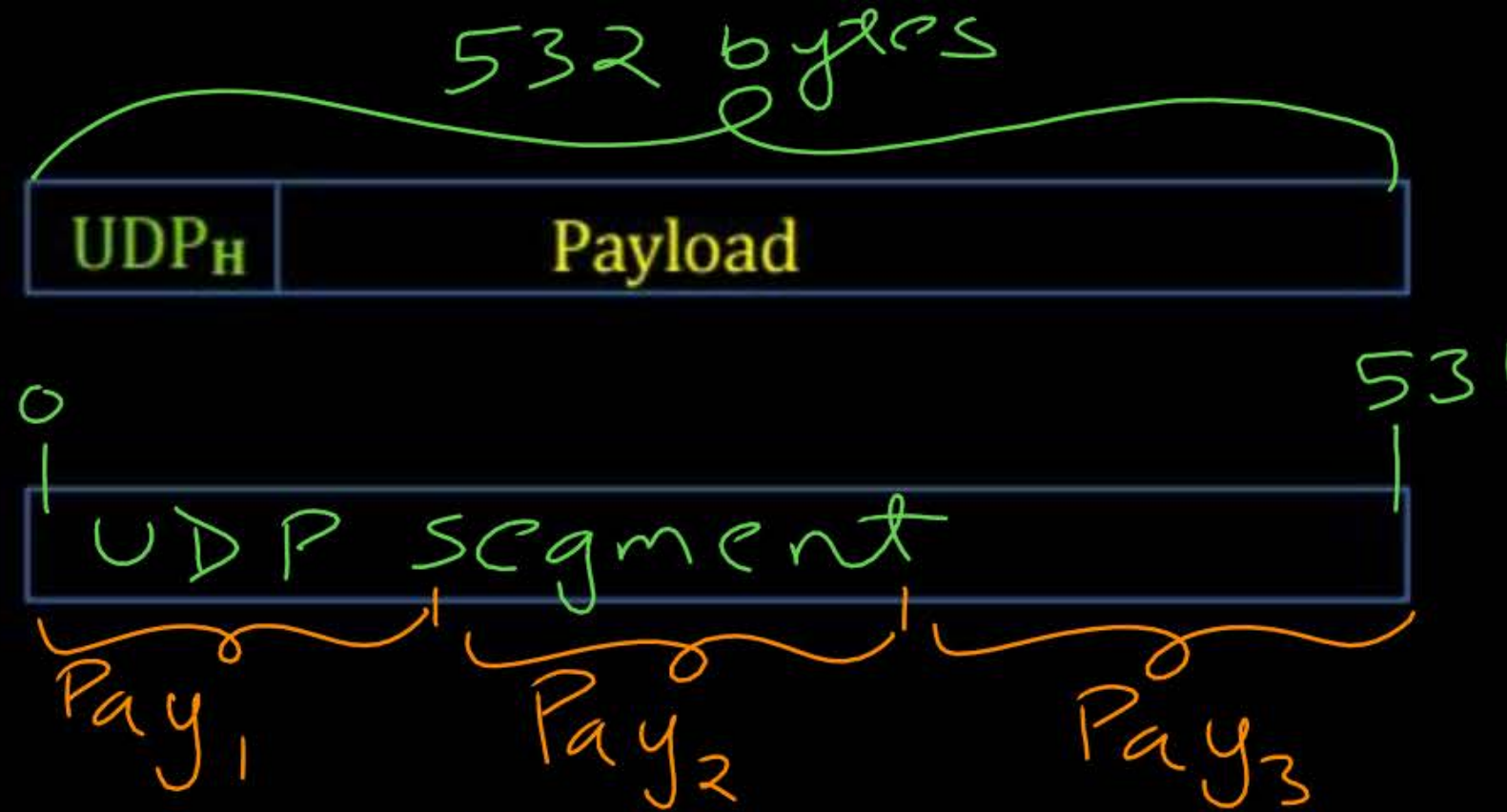
Ans = 3

$$\begin{aligned} \text{MTU} &= [200 \text{ bytes}] \\ \text{IPv4 Header Size} &= [20 \text{ bytes}] \end{aligned}$$

$$\begin{aligned} \text{Maximum Payload Size} &= [\text{MTU} - \text{Header Size}] \text{ bytes} \\ &= (200 - 20) \text{ bytes} \\ &= 180 \text{ bytes} \\ &\approx 176 \text{ bytes (when } M=1) \end{aligned}$$

$$\text{UDP Segment Size} = 532 \text{ bytes}$$

$$\begin{aligned} \text{Total Number of IP fragments [N]} &= \left\lceil \frac{\text{UDP Segment Size}}{\text{Payload Size}} \right\rceil \\ &= \left\lceil \frac{532 \text{ bytes}}{176 \text{ bytes}} \right\rceil \\ &= \lceil 3.02 \rceil \end{aligned}$$

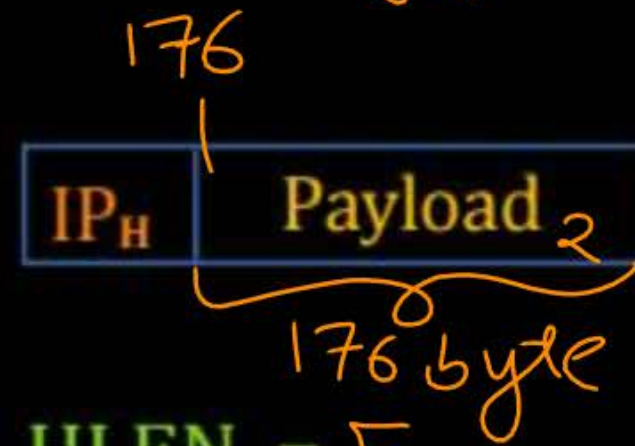


HLEN = 5

TL = 196

Offset = 0

MF bit = 1

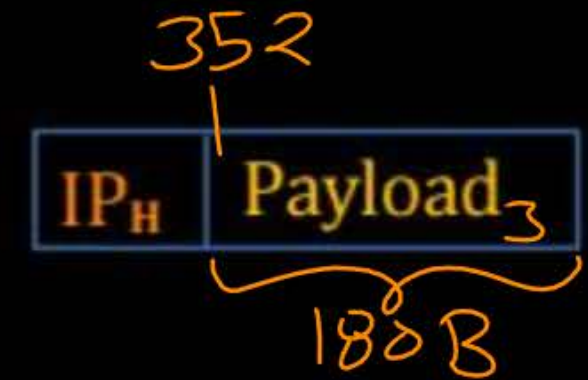


HLEN = 5

TL = 196

Offset = 22

MF bit = 1



HLEN = 5

TL = 200

Offset = 44

MF bit = 0

Example 13 :-

[NAT]



#Q. Consider an IPv4 datagram, where values in total length field and HLEN field are 500 and 10 respectively. Determine the value of more fragment (MF) flag?

Ans = 0

$$\text{HLEN} = [10]$$

$$\text{Total Length} = [500]$$

$$\begin{aligned} \text{Old Payload Size} &= [\text{Total Length} - (\text{HLEN} * 4)] \text{ bytes} \\ &= [500 - (10 * 4)] \text{ bytes} \\ &= 460 \text{ bytes} \end{aligned}$$

- * The payload size in given IPv4 Packet is not multiple of 8 bytes.
- * Definitely this IP packet is last fragment in the sequence. [M bit should be zero]

Example 14 :-

[NAT]



#Q. Consider an IPv4 datagram, where values in total length field and HLEN field are 500 and 5 respectively. Determine the value of more fragment (MF) flag?

Ans = Can't
determine

= 0 or 1

$$\text{HLEN} = [5]$$

$$\text{Total Length} = [500]$$

$$\begin{aligned} \text{Old Payload Size} &= [\text{Total Length} - (\text{HLEN} * 4)] \text{ bytes} \\ &= [500 - (5 * 4)] \text{ bytes} \\ &= 480 \text{ bytes} \end{aligned}$$

* Payload size in given IP packet is multiple of 8 bytes.



Topic : Fragmentation Offset



→ For all IPv4 intermediate fragments (except last, those have MF Flag = 1), payload size should be in words [multiple of 8 bytes]

→ For IPv4 last fragment in the sequence (MF Flag = 0), No any restriction on payload size

→ The IPv4 fragment in which offset value is "Zero", is the first fragment in the sequence



Topic : Reassembly of Fragments



→ Reassembly of fragments, only at destination host

→ if any of the fragment is missing (or lost) in the sequence,
this may lead reassembly failure



IP : connection less
(V₄ & V₆) Unreliable
best effort delivery
packet switched network Protocol

* No any ACK for IP packets

[MSQ]

H.W.

ISC

[GATE-2024][1 Mark]



#Q. Which of the following statements about IPv4 fragmentation is/are TRUE?

- A** The fragmentation of an IP datagram is performed only at the source of the datagram
- B** The fragmentation of an IP datagram is performed at any IP router which finds that the size of the datagram to be transmitted exceeds the MTU
- C** The reassembly of fragments is performed only at the destination of the datagram
- D** The reassembly of fragments is performed at all intermediate routers along the path from the source to the destination

[MSQ] H.W.

11T-B

[GATE-2021][2 Mark]



#Q. Consider two hosts P and Q connected through a router R. The maximum transfer unit (MTU) value of the link between P and R is 1500 bytes, and between R and Q is 820 bytes.

A TCP segment of size 1400 bytes was transferred from P to Q through R, with IP identification value as 0x1234. Assume that the IP header size is 20 bytes. Further, the packet is allowed to be fragmented, i.e., Don't Fragment (DF) flag in the IP header is not set by P.

Which of the following statements is/are correct?

- ☐ A Two fragments are created at R and the IP datagram size carrying the second fragment is 620 bytes.
- ☐ B If the second fragment is lost, R will resend the fragment with the IP identification value 0x1234.
- ☐ C If the second fragment is lost, P is required to resend the whole TCP segment.
- ☐ D TCP destination port can be determined by analysing only the second fragment.



Topic : Fragmentation Offset



- Total Length field size = 16 bits
- Maximum Ipv4 datagram size = $[2^{16} - 1]$ bytes

Max^m $(2^{16} - 1)$ bytes



20 B
Base
(Min^m) Header

Max^m $(2^{16} - 21)$ bytes

$(2^{16} - 22) \Rightarrow 16 \text{ bit field}$



Topic : Fragmentation Offset



→ Fragmentation Offset field size = 13-bits

=> Why fragmentation offset is in word (of 8 bytes)?

→ Fragmentation may be needed at any intermediate router

→ Solution : Fragmentation Offset in Words [word size = 8 bytes]





$$8 \rightarrow 1000$$

$$16 \rightarrow 10000$$

$$24 \rightarrow 11000$$

$$32 \rightarrow 100000$$

$$40 \rightarrow 101000$$

$$\frac{40}{8} = 5$$

$$40 \rightarrow 101000$$

$$5 \rightarrow \boxed{101}$$

$$5 \times 8 \rightarrow 101000$$



Topic : DF Flag



DF : Do not fragment

- Source host can restrict further fragmentation (division) of datagram at any intermediate router *during routing*
- In such cases, source host set DF bit [DF = 1]
- e.g. "Remote Booting"

- Default value of DF flag is "Zero"
- Set / reset by source host only.



Topic : DF Flag



At intermediate Router :- [For arrived IP Packet]

if [DF_bit == 1] AND [Total Length > next network MTU]

- Router discard the IP packet
- generate ICMP Error Message for source host
“Fragmentation Needed and Don't Fragment was Set”



2 mins Summary



Topic

Fragmentation & Reassembly

TTL
Protocol Type
Header checksum

Options
ICMP



THANK - YOU

